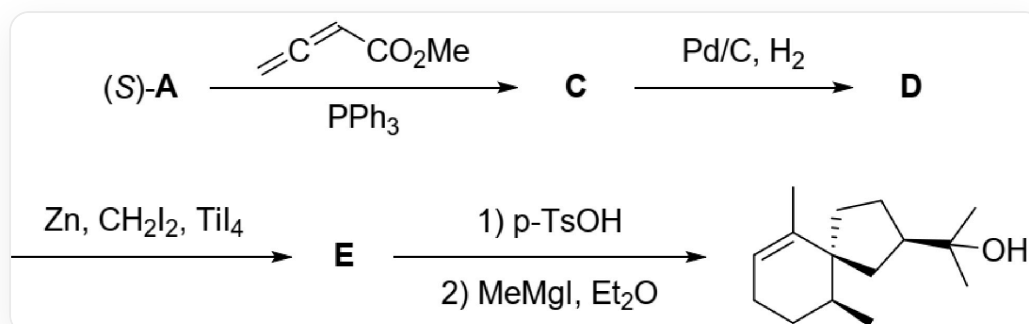


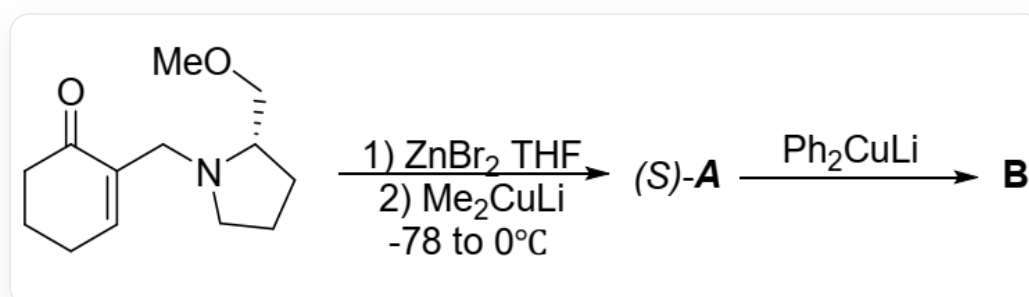
Question

(*S*) – **A** can utilize the [3+2] reaction to construct the basic skeleton of the compound. Pay attention to the stereochemistry of each substance.



(*S*) – **A** reacts with triphenylphosphine and $\text{C}=\text{C}=\text{CC}(\text{OC})=\text{O}$ to obtain **C**, **C** is hydrogenated under palladium on carbon catalysis to obtain **D**, **D** reacts with zinc, diiodomethane and titanium tetraiodide to obtain **E**, **E** is first reacted with *p* – *TsOH*, and then with methyl Grignard reagent and diethyl ether to obtain C[C@H]1CCCC=C(C)[C@@]12C[C@H](C(O)(C)C)CC2

(*S*) – **A** can be prepared by the following reaction and has the following reactions:



O=C1C(CN2CCC[C@H]2COC)=CCCC1 is first reacted with zinc bromide in *THF*, then reacted with lithium dimethylcuprate, the system temperature is raised from minus 78 degrees Celsius to zero degrees to obtain (*S*) – **A**, then (*S*) – **A** reacts with lithium diphenylcuprate to obtain **B**

Choose the correct option from the following

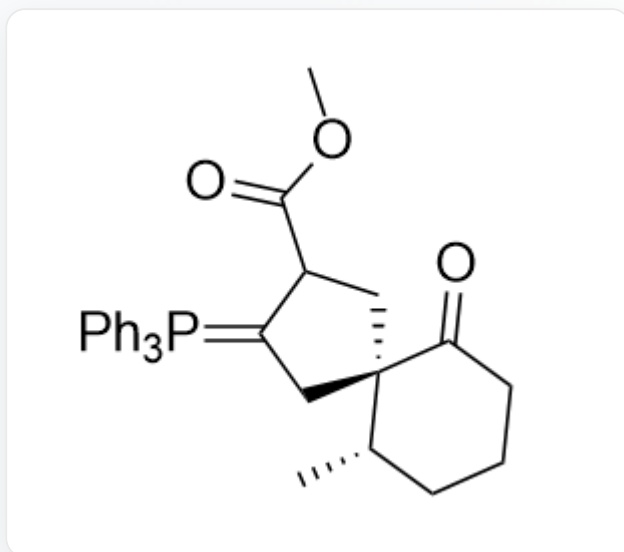
A. All other options are incorrect

B. The newly generated chiral center in **B** has an absolute configuration of S.

C. The most stable conformation of **B** has an axial bond on the six-membered ring.

D. Generating (*S*) – **A** can achieve the goal of controlling chirality in one step without adding zinc bromide.

E. (*S*) – **A** generating **C** involves an intermediate with the structure



O=C1[C@@]2(CC(C(OC)=O)C(C2)=P(C3=CC=CC=C3)(C4=CC=CC=C4)C5=CC=CC=C5)[C@@H](C)CCC1

F. The chemical formula of **E** is $C_{14}H_{20}O_3$

G. **E** The two-step reaction sequence to obtain the product can be interchanged.

Answer

Correct Answer: **E**

Detailed Explanation

O=C1C(CN2CCC[C@H]2COC)=CCCC1 can form a complex with zinc bromide CO1C[C@@H]2CCC[N@@]23CC4=CCCCC4=O[Zn@]13[R] (R represents other possible ligands), fixing the chiral environment near the nucleophilic site. Subsequently, lithium dimethylcuprate nucleophilically attacks from the less hindered side and the chiral auxiliary leaves to obtain (*S*) – **A**, with the structure C=C1[C@@H](C)CCCC1=O.

CHECKPOINT

1 PTS

The formation of (*S*) – **A** requires the participation of zinc bromide to form a complex that fixes the chiral environment

After the addition of lithium diphenylcuprate, an enolate anion is generated, followed by workup to produce a new carbonyl group. This process minimizes the repulsion of all substituents on the six-membered ring, so they are all in equatorial positions. The structure of **B** is O=C1[C@H](CC2=CC=CC=C2)[C@@H](C)CCC1

CHECKPOINT

1 PTS

The structure of **B** is O=C1[C@H](CC2=CC=CC=C2)[C@@H](C)CCC1, all in equatorial positions

From the structure of **B**, it can be seen that the absolute configuration of the newly generated chiral center is R

CHECKPOINT

1 PTS

The absolute configuration of the newly generated chiral center is R

(*S*) – **A** reacts with triphenylphosphine and $C=C=CC(OC)=O$ to obtain **C**. First, triphenylphosphine adds to $C=C=CC(OC)=O$ to form a zwitterion $C=C([CH-]C(OC)=O)[P+](C1=CC=CC=C1)(C2=CC=CC=C2)C3=CC=CC=C3$, followed by a [3+2] reaction, reacting from the less hindered side to produce another intermediate $O=C1[C@@]2(CC(C(OC)=O)C(C2)=P(C3=CC=CC=C3)(C4=CC=CC=C4)C5=CC=CC=C5)[C@@H](C)CCC1$. Finally, the negative charge is transferred to the α -position of the ester group, eliminating triphenylphosphine to obtain **C**, with the structure $C[C@H]([C@]12CC(C(OC)=O)=CC2)CCCC1=O$

CHECKPOINT

1 PTS

The formation of **C** involves the intermediate $O=C1[C@@]2(CC(C(OC)=O)C(C2)=P(C3=CC=CC=C3)(C4=CC=CC=C4)C5=CC=CC=C5)[C@@H](C)CCC1$

C is hydrogenated over palladium on carbon from the less hindered side to obtain **D**, with the structure $C[C@H]([C@]12C[C@H](C(OC)=O)CC2)CCCC1=O$

The reaction of **D** to obtain **E** is the Lombardo reaction. First, zinc reduces diiodomethane to obtain methylene carbene. The methylene carbene reacts with titanium tetraiodide to obtain $C=[Ti](I)I$, which reacts with the carbonyl group to produce $C[C@H]1CCCC2(O[Ti](I)(I)C2)[C@@]13C[C@H](C(OC)=O)CC3$, eliminating $O=[Ti](I)I$ to obtain **E**, with the structure $C[C@H]([C@]12C[C@H](C(OC)=O)CC2)CCCC1=C$, and the chemical formula $C_{14}H_{22}O_2$

CHECKPOINT

1 PTS

The chemical formula of **E** is $C_{14}H_{22}O_2$

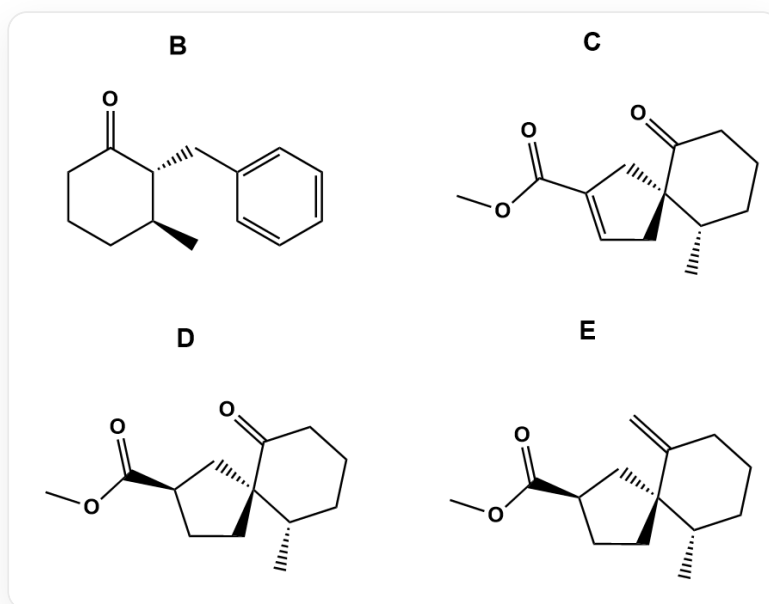
E reacts with $p-TsOH$, and the double bond undergoes isomerization, yielding C[C@H]1CCC=C(C)[C@@]12C[C@H](C(OC)=O)CC2. Subsequently, methyl Grignard reagent reacts with the ester group, adding twice to obtain the product. If the Grignard reagent is used to react with **E** first, the ester group can be smoothly converted to a tertiary alcohol, but the hydroxyl group may be difficult to retain after treatment with $p-TsOH$, so the order of these two reaction steps cannot be reversed.

CHECKPOINT

1 PTS

Reversing the order of the two reaction steps to obtain the product from **E** may lead to the tertiary alcohol being difficult to retain

Therefore, choose option E



本题中结构